

Convex Optimization Review Slides

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August 14, 2025

- EE364A student-run final exam review.
- Focus: selected problems from HW to prepare for final exam.
- Goal: refresh key concepts and walk through representative problem structures.

Overview

- 1 Introduction
- 2 HW #1
- 3 HW #3
- 4 Acknowledgements

HW #1 — Key Concepts

- What is an optimization problem?
- Introduction to convex sets
- Determining if a set is convex

Optimization Problem

An optimization problem is a problem of the form:

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m, \\ & g_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- f_0 is the **objective function**
- x is the variable to be chosen
- f_i are inequality constraint functions $\forall i \in \{1, \dots, m\}$
- g_i are equality constraint functions $\forall i \in \{1, \dots, p\}$

Several variations exist, we are not limited to minimizing one objective functions. Some examples are: maximizing objective functions and having multiple objective functions.

Definition: Convex Set

A set C is convex if the line segment between any two points in C , lies in C . In other words, for any $x_1, x_2 \in C$ and any θ such that $0 \leq \theta \leq 1$, we have

$$\theta x_1 + (1 - \theta)x_2 \in C$$

examples (one convex, two nonconvex sets)

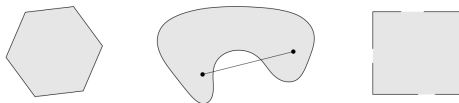


Figure 1: Convex Sets

Examples of Convex Sets

We encountered several examples of convex sets including but not limited to:

- Hyperplanes and Halfspaces
- Cones
- Convex Hulls
- Ellipsoids

Key problem: Determining if something is convex.

Problem A1.1

Determine if the following statements are **true** or **false**.

- (a) Least Squares is a special type of convex optimization.
- (b) By and large, convex optimization problems can be solved efficiently.
- (c) Almost any problem you'd like to solve in practice is convex.
- (d) Convex optimization problems are attractive because they always have a unique solution.

Problem A1.2

A1.2 *Device sizing.* In a device sizing problem the goal is to minimize power consumption subject to the total area not exceeding 50, as well as some timing and manufacturing constraints. Four candidate designs meet the timing and manufacturing constraints, and have power and area listed in the table below.

Design	Power	Area
A	10	50
B	8	55
C	10	45
D	11	50

Are the statements below true or false?

- (a) Design B is better than design A.
- (b) Design C is better than design A.
- (c) Design D cannot be optimal.

HW #3 — Key Concepts

- Understanding convex optimization problems geometrically.
- Find solutions to optimization problems in Python using **CVXPY**.
- Express problems in Disciplined Convex Programming (DCP) form.

Oftentimes for some optimization problems in $\mathbb{R}^2$ and $\mathbb{R}^3$, we can have good intuition for the solution of the problem using its geometric structure

We will look at **Problems 4.1 and A4.2** to get an understanding.

Problem 4.1

Consider the optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x_1, x_2) \\ \text{subject to} & 2x_1 + x_2 \geq 1, \\ & x_1 + 3x_2 \geq 1, \\ & x_1, x_2 \geq 0\end{array}$$

Make a sketch of the feasible set. For the following objective functions, give the optimal set and optimal value.

- $f_0(x_1, x_2) = x_1 + x_2$
- $f_0(x_1, x_2) = \max\{x_1, x_2\}$
- $f_0(x_1, x_2) = x_1^2 + 9x_2^2$

Problem 4.1

For $f_0(x) = x_1 + x_2$ as the objective function, we have the feasible set:

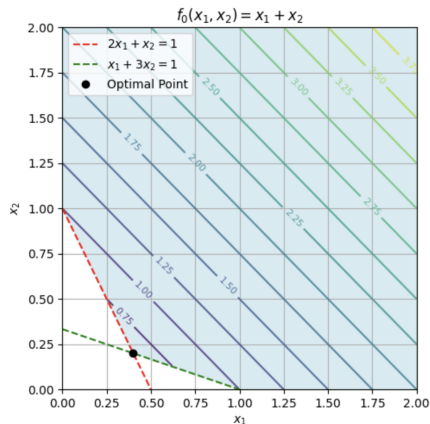


Figure 2: $f_0(x) = x_1 + x_2$ Plot

Problem 4.1

For $f_0(x_1, x_2) = \max\{x_1, x_2\}$ as the objective function, we have the feasible set:

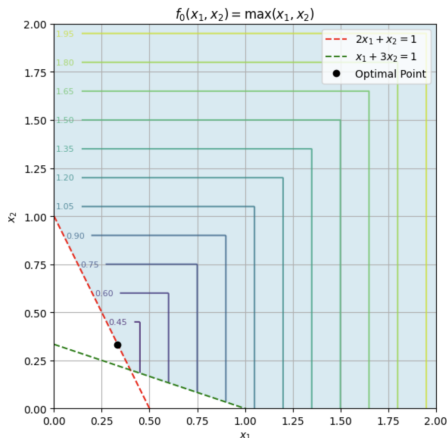


Figure 3: $f_0(x) = \max\{x_1, x_2\}$ Plot

Problem 4.1

For $f_0(x_1, x_2) = x_1^2 + 9x_2^2$ as the objective function, we have the feasible set:

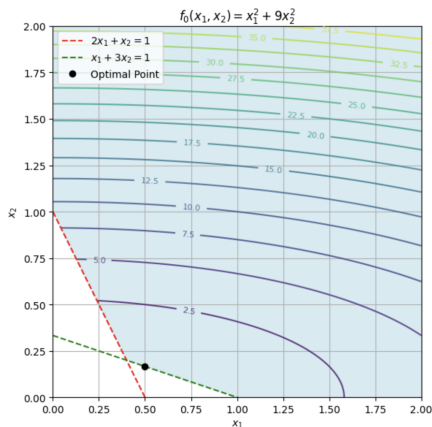


Figure 4: $f_0(x) = x_1^2 + 9x_2^2$ Plot

Problem A4.2

We were able to gain a rough idea of what the objective function, constraint functions, and feasible set look like. However what we have greater need for is to find exact values for the optimal value of the optimization problem.

We find these exact values using *CVXPY*.

Disciplined Convex Programming (DCP)

- DCP is a given set of rules to ensure a problem is convex when written properly
- It exists to ensure that solvers are not passed non-convex formulations for convex solvers. The solver will alert you with an error!
- Every expression passed into the problem must follow allowed composition patterns

Problem A4.25

Probability bounds. Consider random variables X_1, X_2, X_3, X_4 that take values in $\{0, 1\}$. We are given the following marginal and conditional probabilities:

$$\begin{aligned}\mathbf{prob}(X_1 = 1) &= 0.9, \\ \mathbf{prob}(X_2 = 1) &= 0.9, \\ \mathbf{prob}(X_3 = 1) &= 0.1, \\ \mathbf{prob}(X_1 = 1, X_4 = 0 \mid X_3 = 1) &= 0.7, \\ \mathbf{prob}(X_4 = 1 \mid X_2 = 1, X_3 = 0) &= 0.6.\end{aligned}$$

Explain how to find the minimum and maximum possible values of $\mathbf{prob}(X_4 = 1)$, over all (joint) probability distributions consistent with the given data. Find these values and report them.

Hint: In CVXPY, you can create a multidimensional array. For example, `cp.Variable((2, 2, 2, 2))` constructs a 4-dimensional array of variables, with each of the four indices taking the values 0 or 1. The function `cp.sum` takes an axis argument that allows you to sum along dimensions selectively. For example, `cp.sum(x, axis=(0, 3))` creates a $(2, 2)$ variable by summing along the first and last dimensions.

Acknowledgements

This presentation was made possible thanks to:

- Daniel Cederberg (Stanford University)
- Teaching team for EE364A (Summer 2025)
- *Convex Optimization* — Boyd and Vandenberghe
- *Additional exercises for Convex Optimization* - Boyd and Vandenberghe
- Creators and maintainers of CVXPY
- Rohit Makhija for the opportunity to present
- All my classmates in EE364A

Thank you!